

ECE3441 – FALL 2025

PRACTICE PROBLEMS

Solve the following Problems from your Textbook

CHAPTER 13

Problems 13.1, 13.4, 13.7, 13.11, 13.13,

1) Problem 13.1 - Solutions in Textbook

2) Problem 13.4 - Solutions in Textbook

3) Problem 13.7

Solution

$$Q_1^+ = J_1 Q_1' + K_1' Q_1$$

$$Q_1^+ = X Q_1' + X Q_2' Q_1$$

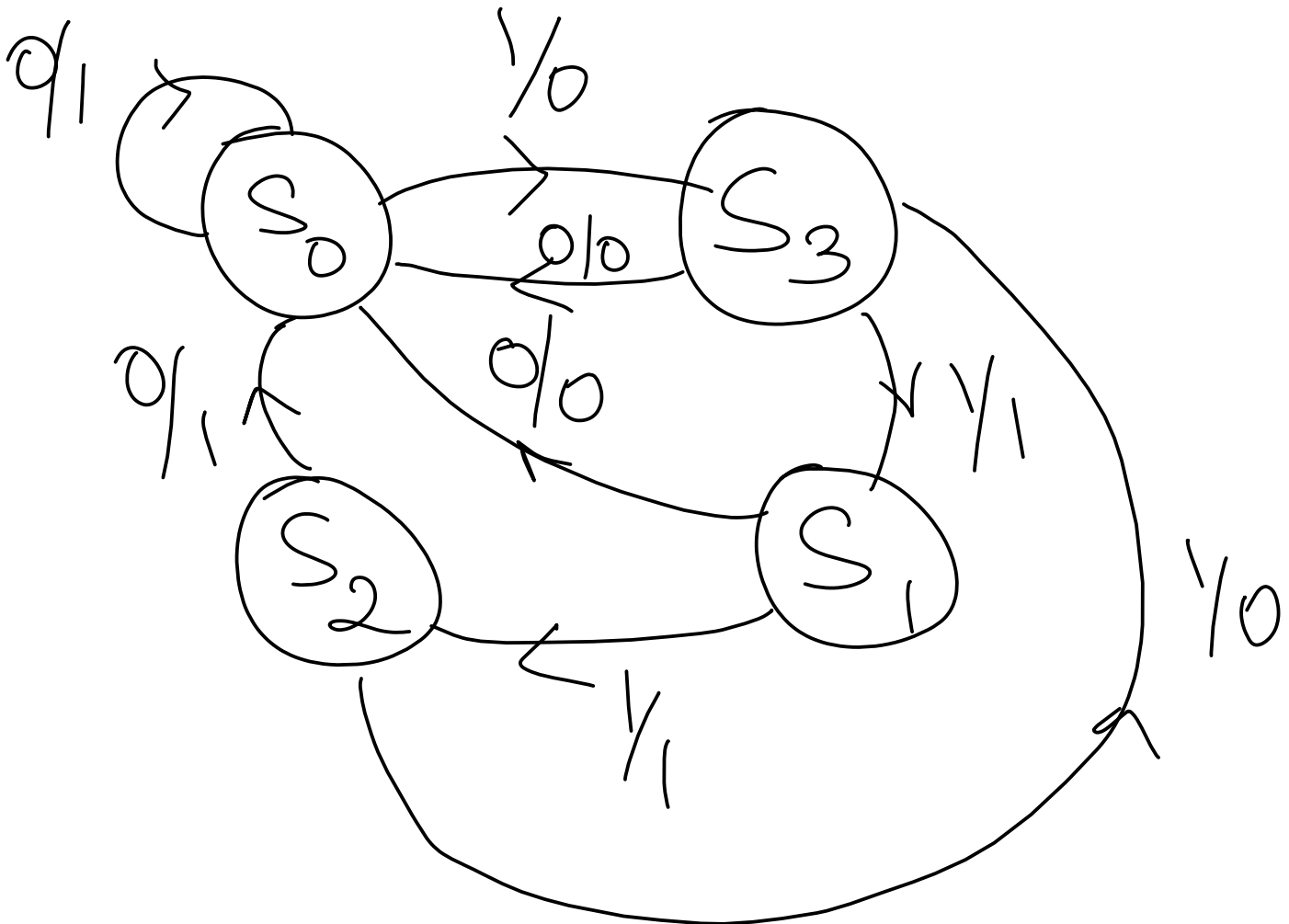
$$\begin{aligned} Q_2^+ &= J_2 Q_2' + K_2' Q_2 \\ &= X Q_2' + X Q_1 Q_2 \end{aligned}$$

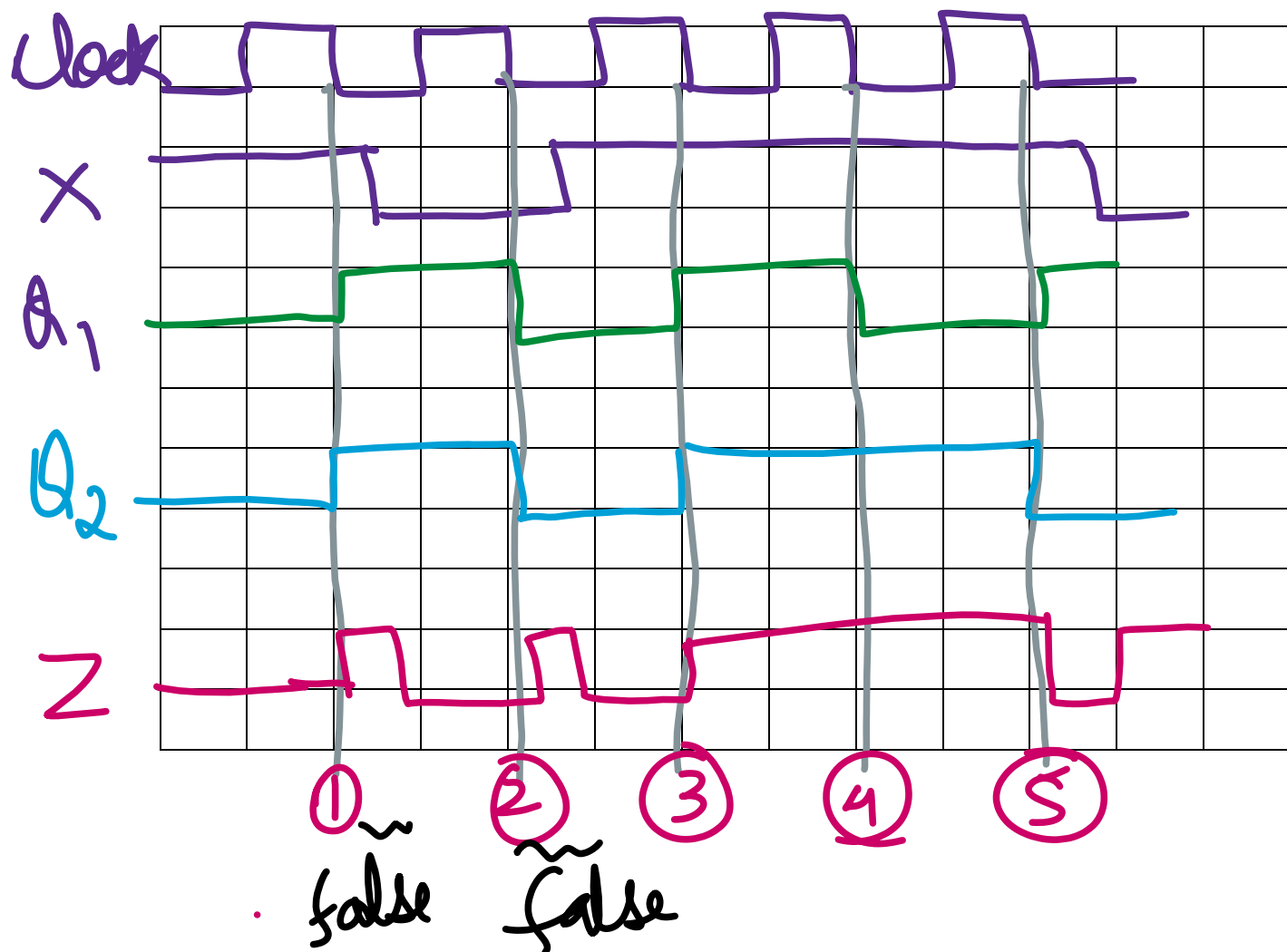
$$Z = X Q_2' + X Q_2$$

Transition Table

Present State Q1 Q2	Next State Q1 ⁺ Q2 ⁺		Z	
	X=0	X=1	X=0	X=1
00	00	11	1	0
01	00	10	0	1
11	00	01	0	1
10	00	11	1	0

State Graph





c) $z = 00011$

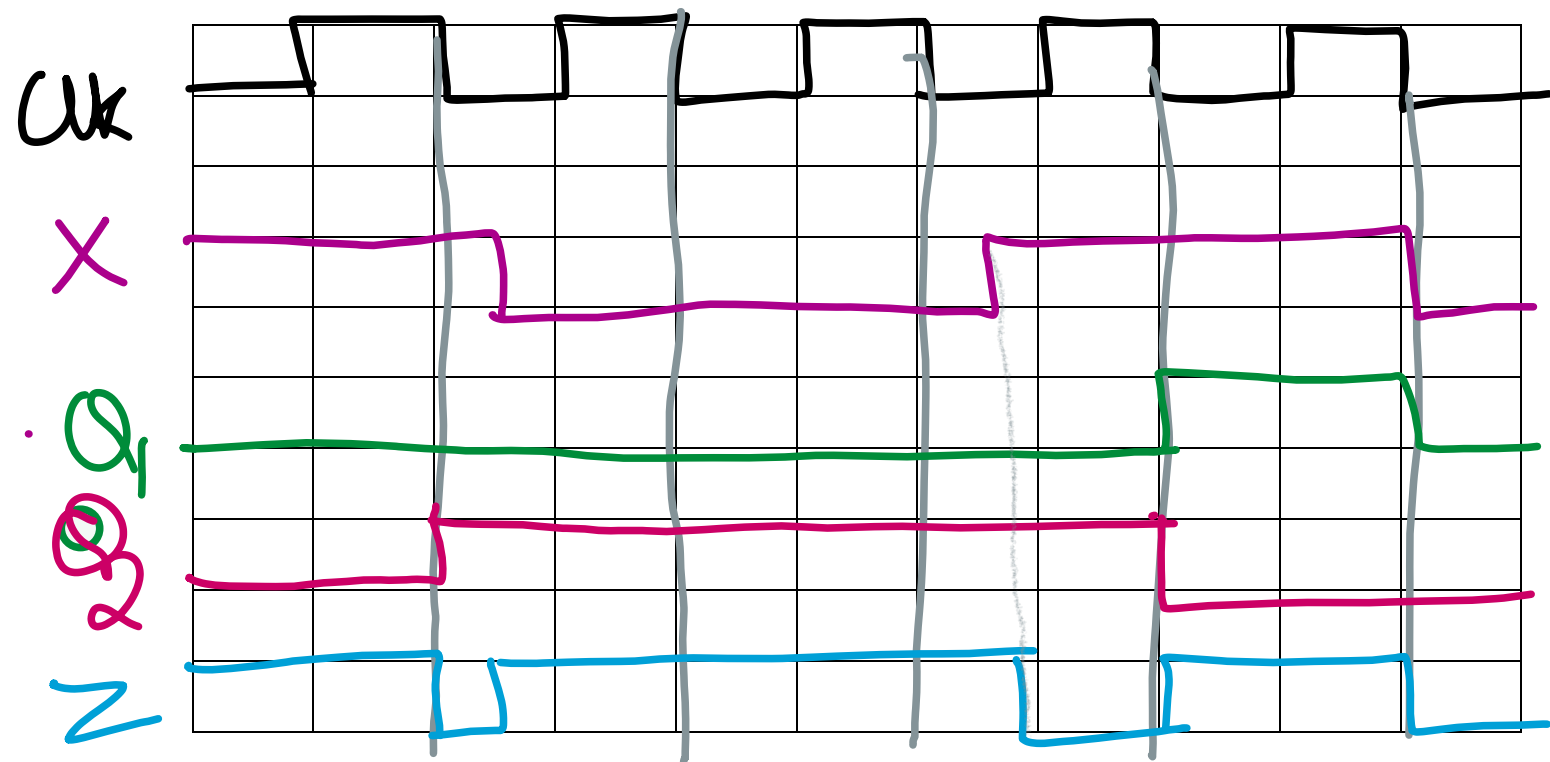
Question 13.11

Z depends on X , so it is a Mealy Machine

$$\begin{aligned}Q_1^+ &= J_1 Q_1' + K_1' Q_1 = X Q_1' Q_2 + X' Q_1 \\Q_2^+ &= J_2 Q_2' + K_2' Q_2 = X Q_1' Q_2' + X' Q_2 \\Z &= X Q_2' + X' Q_2\end{aligned}$$

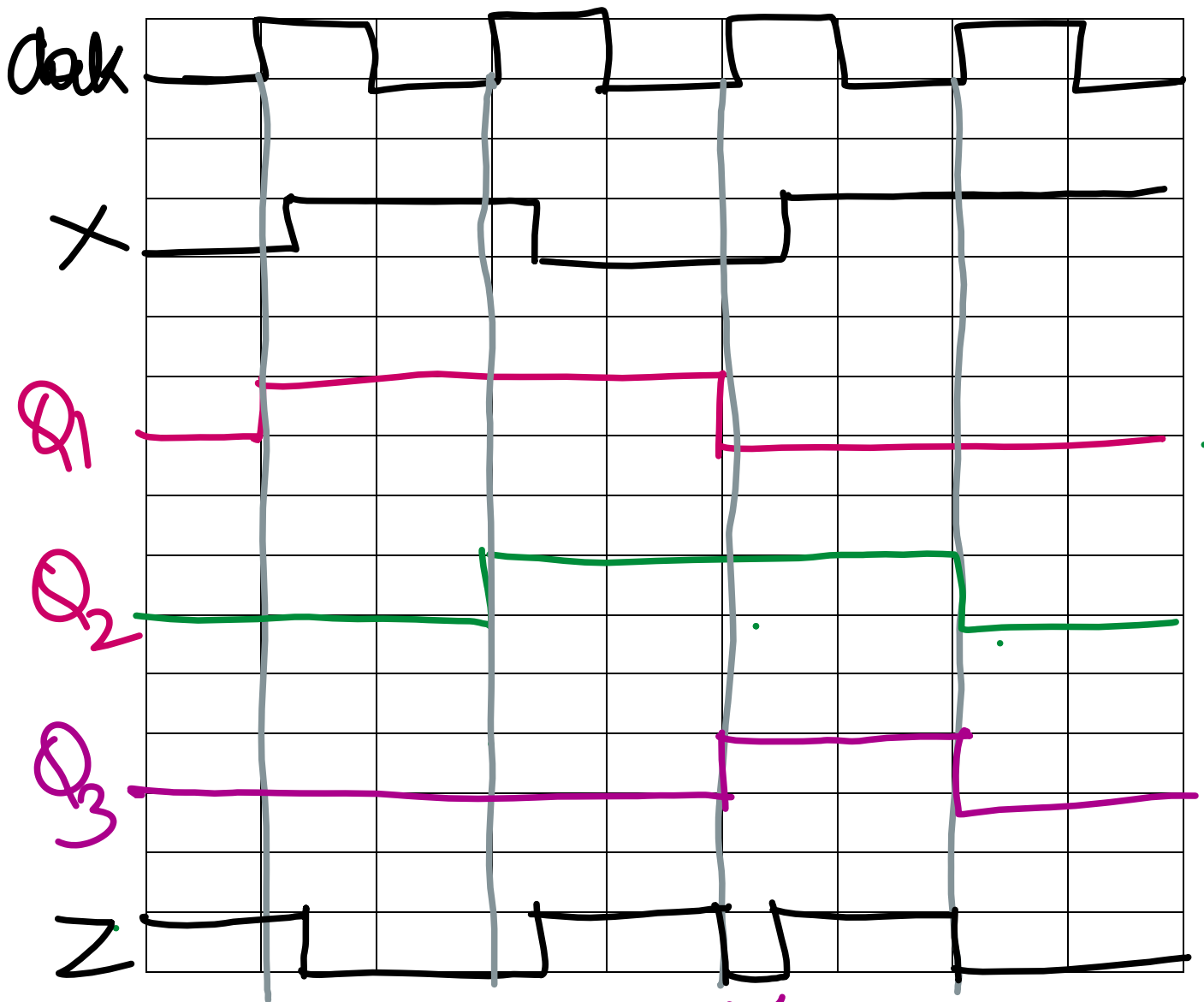
Transition Table

Present State Q1 Q2	Next State Q1 ⁺ Q2 ⁺		Z	
	X=0	X=1	X=0	X=1
00	00	01	0	1
01	01	10	1	0
11	11	00	1	0
10	10	00	0	1



~ false output

Question 13.13



Correct output - 1011

false